

Resto AI Agent - Project Proposal

1. Team

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2. Company

Deepwisdom & StepFun

The project can leverage DeepWisdom's AI-agent capabilities in Atoms and StepFun's large language models to build a practical, agentic solution for small restaurant operations.

3. Business: what products, customers, and use cases does the company already have?

Products/Capabilities

- Large language models for natural-language understanding and generation.
- AI-agent capabilities for reasoning, tool use, and multi-step task execution.
- AI capabilities that can interact with external data and business tools.

Target Customers:

Small and medium-sized restaurant owners and operators, particularly those who:

- Manage operations themselves or with a small team.
- Still rely heavily on paper invoices, WhatsApp messages, spreadsheets, or manual records.
- Have limited time and technical resources for business analysis.
- Need help managing procurement, inventory, payments, and digital marketing.

Relevant Use Cases

- Understanding and digitizing invoices and receipts.
- Tracking inventory and purchasing.
- Comparing supplier prices and identifying price changes.
- Monitoring supplier payments and cash flow.
- Summarizing restaurant performance.
- Supporting business decisions through natural-language conversations.
- Generating marketing ideas and content for social media.

4. Value: what can we build that creates real value for the company and users?

For **DeepWisdom**, this project provides an opportunity to apply the Atoms platform to build and experiment with a practical AI agent, while using **StepFun's** models to evaluate their capability in handling real-world, long-horizon tasks involving multiple steps and continuous decisions.

For **restaurant owners**, the agent reduces repetitive manual work in areas such as invoice processing, inventory tracking, and daily operations, helping them save time and energy while making their business easier to manage.

5. Your Goal: What do you want to gain from this project — portfolio, skills, users, or impact?

Build and validate a practical AI-agent operating assistant for small restaurants that reduces the manual work required to manage procurement, inventory, cash flow, and marketing.

- **Skills:** Gain hands-on experience with LLMs, multimodal document understanding, voice interfaces, agent workflows, tool calling, structured data, RAG, and business analytics.
- **Impact:** Help owners save time and make better purchasing and operational decisions.
- **Business Understanding:** Explore how AI agents can move from answering questions to continuously monitoring operations and recommending or executing actions.

6. Action Items: what can each person do this week to move the project forward?

In the following week, the team will first talk with restaurant owners to understand their daily operational pain points and identify the most repetitive tasks worth automating. Based on these findings, we will prioritize and build the first core agent, focusing on invoice processing, inventory tracking, and operational summaries. We will then test the agent with realistic restaurant data and scenarios, evaluate its accuracy, usefulness, and ability to complete multi-step tasks, and compare the performance of different StepFun models. Based on the results and user feedback, we will iterate on the agent's workflow and capabilities, improve the system through further development and testing, and finally prepare an MVP launch with selected restaurant users for real-world validation.

A. Problems & References

Key Problem Restaurant Owners Face

1. Procurement, Inventory & Supplier Management

Small restaurants often manage purchasing, stock, and suppliers manually through invoices, WhatsApp, spreadsheets, or physical market visits. Without a dedicated procurement team, owners may struggle to track stock levels, detect supplier price changes, compare suppliers, and decide what and when to purchase. This can lead to overstocking, stockouts, and unnecessary costs.

References: BPS reports **5.28 million F&B service businesses in Indonesia in 2024**, indicating a large potential user base. Research on restaurant procurement and Indonesian

F&B MSMEs also highlights the importance of digital platforms and procurement processes for improving operational efficiency and competitiveness.

2. Sales & Menu Management

Small restaurant owners may have sales records but lack the time or technical skills to analyze them using Excel. They need simple answers about best-selling menus, declining sales, ingredient costs, and profitability rather than complex spreadsheets.

References: Research on Indonesian culinary MSMEs shows that digital technology adoption can support business performance and competitiveness, while digital platform capabilities can improve strategic agility in F&B SMEs.

3. Reviews & Customer Insights

Customer feedback is distributed across Google, food delivery platforms, and direct message to Whatsapp. Manually reading reviews makes it difficult to identify recurring complaints, customer preferences, and service problems. Online reviews can significantly influence customer decisions in hospitality.

4. Marketing & Digital Presence

Restaurants need to continuously maintain their digital presence, but small businesses may lack dedicated marketing staff. Owners need help generating content, identifying target customers, planning promotions, and maintaining consistent posting. Beyond regular walk-in customers, restaurants may also need to actively approach new customer segments for catering, private parties, corporate events, and other services, creating additional sources of revenue. An AI agent can help identify potential customers, prepare targeted offers, and manage outreach while maintaining the restaurant's regular digital presence.

Overall Problem:

Small restaurants have fragmented operational and customer data but limited staff and technical expertise. The opportunity is to use AI agents to turn this scattered information into understandable insights, recommendations, and actions rather than requiring owners to manage complex software themselves.

B. Solutions

An AI Restaurant Operations Platform for small and medium-sized Indonesian restaurants, combining a WhatsApp AI agent with a simple web dashboard. Through WhatsApp, owners can simply send invoice/receipt photos, voice notes, or text messages. The AI agents process this information and help manage procurement, suppliers, sales, customers, and marketing through AI agents.

The web dashboard provides a more detailed view of restaurant operations through key metrics and interactive visualizations, including sales trends, inventory levels, supplier prices, customer sentiment, pending supplier payments, and marketing performance. This allows owners to monitor their business without relying on complex spreadsheets or manual analysis.

Below is the details use case of the AI agent:

1. **Procurement & Inventory Agent**

Manage purchases, invoices, stock, and reorder decisions.

Invoice → Purchase → Inventory → Reorder

- a. Extract invoice
- b. Create purchase record
- c. Update inventory
- d. Detect low stock
- e. Calculate reorder quantity
- f. Compare historical prices
- g. Generate purchase recommendation
- h. Ask for owner confirmation

2. **Supplier Intelligence Agent**

Find suppliers, compare prices, monitor price changes, and recommend suppliers.

Supplier discovery → Price comparison → Supplier evaluation → Recommendation

- a. Search supplier
- b. Register supplier
- c. Compare supplier prices
- d. Track historical prices
- e. Detect price increases
- f. Evaluate supplier reliability
- g. Recommend alternatives
- h. Estimate potential savings

3. **Sales & Menu Intelligence Agent**

Analyze sales, menu performance, costs, and trends.

Sales data → Trend analysis → Menu performance → Business insights

- a. Import sales data
- b. Analyze sales trends
- c. Rank menu items
- d. Analyze menu profitability
- e. Detect unusual sales changes
- f. Extract keywords, categorize sentiment, and map issues to menu items or staff changes
- g. Identify popular combinations
- h. Recommend menu actions

4. **Customer Service Agent**

The agent can proactively engage customers after a visit or order, collect feedback, identify complaints, answer common questions, and escalate serious issues to the owner.

Customer interaction → Understand → Respond / Collect feedback → Detect complaint → Report / Escalate

- a. Identify customers to follow up with after a visit or order
- b. Proactively ask customers about their experience
- c. Collect customer feedback through messages, reviews, and social media
- d. Detect complaints and negative sentiment
- e. Categorize complaints by issue, such as food quality, service, price, or cleanliness
- f. Identify recurring complaints and emerging problems
- g. Summarize customer feedback and key issues for the owner
- h. Alert the owner when serious or repeated complaints are detected

5. Marketing Agent

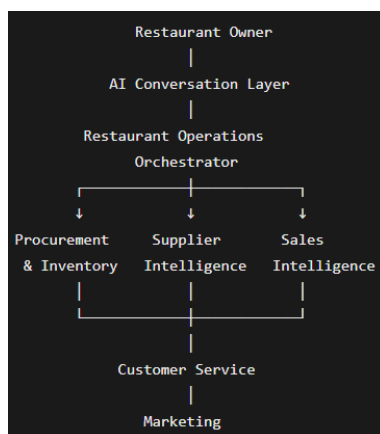
Manage digital presence, content ideas, campaigns, and customer targeting.

Restaurant data → Customer targeting → Content and Campaign planning

- a. Analyze existing and potential customer segments
- b. Identify potential catering, party, and corporate clients
- c. Find relevant target businesses or customer groups
- d. Generate targeted promotional offers
- e. Generate social media content ideas and captions
- f. Create content and campaign schedules
- g. Recommend suitable channels and outreach timing
- h. Track campaign and outreach performance
- i. Analyze customer responses and engagement
- j. Recommend follow-up actions and new opportunities

And then have a **Restaurant Operations Orchestrator** above them that coordinates the agents and provides the owner with alerts, recommendations, and daily summaries.

- a. Detect important events
- b. Trigger the appropriate agent
- c. Combine information from multiple agents
- d. Send alerts
- e. Generate daily summaries
- f. Prioritize recommended actions



How does our solution stand out from competitors?

Unlike traditional restaurant POS platforms (such as Moka) or complex back-office systems (such as MarketMan and ZERLO) that require structured data entry and rigid workflows, our solution functions as an autonomous, conversational operational layer. Instead of forcing busy owners to adopt another all-in-one ERP, our multi-agent architecture ingests messy, unstructured real-world inputs, such as WhatsApp voice notes, forwarded supplier chat messages, and receipt photos, and processes them without manual setup. Furthermore, while broader marketing tools (such as Restomas) focus primarily on guest-facing digitalization, we prioritize back-office operational and supply chain intelligence, providing non-technical owners with proactive execution and actionable recommendations rather than passive dashboards.

C. Assumptions

1. Restaurant owners still rely heavily on manual processes.
Many small and medium-sized restaurants use paper invoices, WhatsApp messages, spreadsheets, and manual records to manage purchases, inventory, suppliers, and payments.
2. Owners have limited time and operational resources.
Restaurant owners often handle multiple responsibilities themselves, so repetitive administrative tasks take significant time and energy away from running and growing the business.
3. The operational data already exists, but is fragmented.
The problem is not necessarily a lack of data. Invoices, receipts, sales records, supplier prices, and customer feedback exist in different formats and places, making them difficult to consolidate and analyze.
4. WhatsApp is a more natural interface than a complex business system.
Owners are more likely to send a photo, voice note, or simple message than manually enter information into a new software system.
5. AI agents can reliably handle repetitive, multi-step operational tasks.
With the right tools, guardrails, and human approval, AI can process documents, update records, analyze trends, and provide useful recommendations instead of only answering questions.
6. Owners need actionable insights, not just dashboards.
The real value is helping owners answer questions such as what should I buy, when should I reorder, which supplier is better, and what needs my attention today?
7. Owners will trust AI if they can remain in control.
The agent should initially focus on recommendations and low-risk automation, while important actions such as payments or supplier orders can require owner confirmation.